EVERY HYPERPLANE IN PROJECTIVE SPACE IS ISOMORPHIC TO PROJECTIVE SPACE OF ONE LOWER DIMENSION WITH THE TWIST MATCHED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four cross-chapter labels (including a definition and a section), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Zero scheme of a section and closed subschemes of projective space as Proj of a graded quotient. Stacks Project, Divisors, Definition divisors-definition-zero-scheme-s (tag 02OQ); Constructions, Lemma constructions-lemma-closed-in-projective-space (tag 01M1). Identify the hyperplane with Proj of the quotient of the polynomial ring by the homogeneous ideal of the section.
- Functoriality of Proj and twists for surjective degree-one generated graded maps. Stacks Project, Constructions, Lemma constructions-lemma-surjective-graded-rings-generated-degree-1-map-proj (tag 01N4). Transfers the graded isomorphism to an isomorphism of Proj with matching twists.

Source reference. Original reference 01XS: sources/stacks-chow/tags.txt, source lines 2531–2531. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GL: sources/stacks-chow/tags.txt, source lines 4532–4532. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OQ: sources/stacks-chow/tags.txt, source lines 3525–3525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01N1: sources/stacks-chow/tags.txt, source lines 2142–2142. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $n \geq 1$. Let $i : H \to \mathbf{P}^n_k$ be a hyperplane. Then there exists an isomorphism*

$$\varphi : \mathbf{P}^{n-1}_k \longrightarrow H$$

such that $i^*\mathcal{O}(1)$ pulls back to $\mathcal{O}(1)$.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. We have $\mathbf{P}^n_k = \text{Proj}(k[T_0, \ldots, T_n])$. The section s corresponds to a homogeneous form in $T_0, \ldots, T_n$ of degree 1, see Cohomology of Schemes, Section 01XS. Say $s = \sum a_i T_i$. Constructions, Lemma 03GL gives that $H = \text{Proj}(k[T_0, \ldots, T_n]/I)$ for the graded ideal I defined by setting I_d equal to the kernel of the map $\Gamma(\mathbf{P}^n_k, \mathcal{O}(d)) \to \Gamma(H, i^*\mathcal{O}(d))$. By our construction of $Z(s)$ in Divisors, Definition 02OQ we see that on $D_+(T_j)$ the ideal of H is generated by $\sum a_i T_i/T_j$ in the polynomial ring $k[T_0/T_j, \ldots, T_n/T_j]$. Thus it is clear that I is the ideal generated by $\sum a_i T_i$. Note that

$$k[T_0, \ldots, T_n]/I = k[T_0, \ldots, T_n]/(\sum a_i T_i) \cong k[S_0, \ldots, S_{n-1}]$$

as graded rings. For example, if $a_n \neq 0$, then mapping S_i equal to the class of T_i works. We obtain the desired isomorphism by functoriality of Proj. Equality of twists of structure sheaves follows for example from Constructions, Lemma 01N1. □